Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100507 | *Subject: Power Systems-I (Apparatus and Modelling)* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Ferranti effect occurs in
(i) Short transmission lines
(ii) Underground cables only
(iii) Lightly loaded long transmission lines
(iv) DC lines

(b) Skin effect is more pronounced in
(i) DC transmission
(ii) Low voltage distribution
(iii) High frequency AC
(iv) Superconducting lines

(c) Series compensation is used to improve
(i) Voltage profile
(ii) Insulation level
(iii) Power transfer capability
(iv) Corona loss

(d) For a long transmission line, the ABCD parameters are used for
(i) Short line approximation
(ii) Lumped parameter only
(iii) Distributed parameter model
(iv) DC lines only

(e) In a Δ-Y transformer, the phase shift is
(i) 0°
(ii) 90°
(iii) 30°
(iv) 180°

(f) Bewley's lattice diagram is used for
(i) Load flow study
(ii) Economic load dispatch
(iii) Travelling wave analysis
(iv) Stability study

(g) The power output of a wind turbine is proportional to
(i) Wind speed
(ii) Square of wind speed
(iii) Cube of wind speed
(iv) Wind speed squared

(h) The main advantage of the per-unit system in power system analysis is
(i) It eliminates the need for transformers
(ii) All per-unit impedances remain unchanged when referred to either side of a transformer
(iii) It increases the power rating
(iv) It reduces the number of equations

(i) Neutral grounding transformers are primarily used to
(i) Increase fault current
(ii) Improve voltage regulation
(iii) Provide a path for zero-sequence current and limit fault current
(iv) Reduce corona loss

(j) Shunt compensation using SVC is primarily used for
(i) Increasing the power transfer capability
(ii) Reducing line resistance
(iii) Dynamic reactive power support and voltage control
(iv) Increasing surge impedance

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Q.2 (a) What are the advantages and challenges of high renewable penetration in power grids? [7]

(b) Derive the expression for maximum power transfer in a short transmission line. [7]

Q.3 (a) What is Surge Impedance Loading? Derive its expression and significance. [7]

(b) A three-phase 400 kV transmission line has resistance R = 0.1 Ω/km, reactance X = 0.4 Ω/km and susceptance Y = 2.5×10^{-6} S/km. For a line length of 200 km, calculate the surge impedance and surge impedance loading. Also find the power transferred when the load angle is 30° (neglect resistance). [7]

Q.4 (a) Describe the steady-state, transient, and sub-transient equivalent circuits of a synchronous machine during short circuit. [7]

(b) A short transmission line has sending end voltage Vs = 220 kV, receiving end voltage Vr = 200 kV, line impedance Z = 20 + j60 Ω. Calculate: [7]
(i) Active and reactive power at receiving end
(ii) Voltage regulation

Q.5 (a) A synchronous generator is connected to an infinite bus. Draw its phasor diagram when operating at lagging power factor. Derive the expressions for real power (P) and reactive power (Q) in terms of terminal voltage (V), excitation voltage (Ef), synchronous reactance (Xs), and power angle (δ). [7]

(b) A synchronous machine is delivering 0.8 pu power at 0.9 pf lagging. The terminal voltage is 1.0 pu. Calculate the excitation voltage (Ef) if Xs = 1.1 pu. Also find the load angle δ. [7]

Q.6 (a) Explain the basic operation of Line Commutated Converter and Voltage Source Converter based HVDC systems. Highlight the major advantages and limitations of each. Also mention one typical application of each. [7]

(b) A surge of 400 kV travels on a line with Zc1 = 400 Ω and enters a cable with Zc2 = 50 Ω. Calculate: (i) Transmitted voltage (ii) Reflected voltage (iii) Reflection and transmission coefficients. [7]

Q.7 (a) Compare Permanent Magnet Synchronous Generator and Induction Generator used in wind energy conversion. [7]

(b) Explain the concept of Insulation Coordination. Describe the different methods used for protection against overvoltages in high voltage systems. [7]

Q.8 (a) Explain the working and purpose of Neutral Grounding Transformers. Why are they preferred for grounding in power systems? Draw the connection diagram. [7]

(b) Explain Series and Shunt compensation of transmission lines with their advantages. [7]

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Q.9 Write short note on ***any two*** of the following: **[7 x 2 = 14]**

(a) Synchronous Grids and Asynchronous (DC) interconnections
(b) Compare Two-winding transformers with auto transformers
(c) Power Electronics interfaces of wind generators to the grid
(d) Distributed Energy Resources & Energy Storage

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